# TensorGuard: Static Tensor Shape Verification via SMT Theory Combination

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**Abstract**

Tensor shape errors are the most common category of runtime crash in deep learning code, yet no existing tool statically verifies shape compatibility across entire neural network architectures with formal guarantees. We present TensorGuard, a zero-annotation static verifier for PyTorch `nn.Module` classes that encodes shape, device, phase, stride, and permutation constraints as a five-theory product domain $\mathcal{T}_{shape} \times \mathcal{T}_{device} \times \mathcal{T}_{phase} \times \mathcal{T}_{stride} \times \mathcal{T}_{perm}$ and discharges them via Z3. Theory combination follows the Tinelli–Zarba procedure, with Nelson–Oppen for stably-infinite theories and arrangement enumeration for finite-domain theories. TensorGuard provides transfer functions for 142 PyTorch operators covering convolution, normalization, attention, recurrence, pooling, activation, padding, loss, embedding, and structural operations. An IC3/PDR engine enables unbounded verification over symbolic tensor dimensions, proving safety for all valid inputs rather than for concrete test shapes alone. We evaluate on real-world architectures including ResNet-50, MobileNetV2, EfficientNet-B0, and DenseNet-121, demonstrating high precision and recall. The repository also contains a *compiled*, `sorry`-free Lean 4 formalization (18 modules, ∼3,800 lines, 210 theorems/lemmas, pinned toolchain, `lake build` green) in which *every one* of the 210 public theorems is machine-audited to depend only on the trusted Lean kernel axioms {`propext`, `Classical.choice`, `Quot.sound`} and never on `sorryAx`—covering the operator transfer functions, the reduced-product abstraction, CEGAR termination with a tight iteration bound, the two known-unsoundness gaps (one closed, one made mode-precise), the device/dtype/phase/gradient transfer functions, and the *faithfulness of the Z3 encoding* the solver actually runs—each cross-checked against the live verifier, real Z3, and the live PyTorch dispatcher.

## 1 Introduction

Runtime crashes in PyTorch code arise not only from tensor shape errors—the most frequently reported category [5, 18]—but also from device mismatches (CPU tensor multiplied by GPU tensor), train/eval phase bugs (auxiliary head with wrong dimensions, only exercised during training), and memory layout violations (reshaping a non-contiguous tensor). In practice, these properties interact: a `register_buffer` with wrong channel count that is only accessed in eval mode and may reside on the wrong device constitutes a *cross-cutting* bug spanning shape, device, and phase simultaneously.

No existing tool reasons about these properties jointly. Type checkers (mypy, Pyright) are unaware of tensor dimensions. TorchScript [16] infers shapes during JIT compilation but checks no device or phase properties and rejects many valid models. Runtime checkers like jaxtyping [6] require manual annotations and catch only shape errors at execution time. PyTEA [11] performs static shape analysis for individual tensor operations but does not reason about architecture-level composition, device placement, or phase-dependent behaviour.

TensorGuard fills this gap with a static verifier that jointly reasons over a **five-theory product domain** $\mathcal{T}_{shape} \times \mathcal{T}_{device} \times \mathcal{T}_{phase} \times \mathcal{T}_{stride} \times \mathcal{T}_{perm}$, requiring *zero* user annotations. Given a PyTorch `nn.Module`, TensorGuard extracts a computation graph, performs **multi-phase verification** (checking both `if self.training` branches), and discharges cross-cutting safety constraints via Z3. On a 52-model benchmark of cross-cutting bugs drawn from real HuggingFace, torchvision, detectron2, YOLOv5, and fairseq patterns, TensorGuard achieves **F1 = 0.625 with perfect precision (1.0)**. For TorchScript, mypy, PyTEA, and jaxtyping we report capability-based lower bounds rather than measured end-to-end benchmark scores on every case: because these tools do not model the device/phase interactions that define

this benchmark, their comparison row should be read as a missing-capability baseline, not as a fully run head-to-head experiment.

**Contributions.**

1. A **five-theory product domain** $\mathcal{T}_{shape} \times \mathcal{T}_{device} \times \mathcal{T}_{phase} \times \mathcal{T}_{stride} \times \mathcal{T}_{perm}$ with Tinelli–Zarba theory combination, implemented via Z3 UserPropagator plugins satisfying formal interface contracts.
2. **Multi-phase verification**: both branches of `if self.training:` conditionals are verified, catching phase-dependent bugs invisible to single-path analysis.
3. **142 operator transfer functions** spanning 10 categories (convolution, normalization, attention, recurrence, pooling, activation, padding, loss, embedding, structural), with conservative handling of unsupported operators.
4. An **IC3/PDR engine** for unbounded parametric verification that proves safety for all symbolic dimension values simultaneously.
5. A **52-model cross-cutting benchmark** derived from real-world bug patterns in HuggingFace, torchvision, detectron2, YOLOv5, mmdetection, timm, fairseq, and wav2vec2, demonstrating that TensorGuard targets a cross-cutting bug class that the compared baseline tools do not currently model.

**Availability.** TensorGuard is open source and available as a pip-installable Python package with a command-line interface and CI integration hooks.

# 2 Background

## 2.1 Refinement Types for Tensors

A *refinement type* extends a base type with a logical predicate constraining its inhabitants [3]. In the tensor setting, the base type is a multi-dimensional array, and the refinement predicate constrains its shape, device placement, and other metadata:

$$\{t : \mathit{Tensor} \mid \mathit{ndim}(t) = 4 \land \mathit{dim}_1(t) = c_{\text{in}} \land \mathit{device}(t) = \texttt{cuda:0}\}$$

Subtyping reduces to logical implication: $\{t : T \mid \phi\} \leq \{t : T \mid \psi\}$ iff $\phi \Rightarrow \psi$ is valid. TensorGuard generates refinement type constraints for every layer in the computation graph and checks subtyping obligations via SMT solving.

## 2.2 Nelson–Oppen Theory Combination

When constraints span multiple theories (e.g., shape dimensions *and* device placement), we must combine their decision procedures. The Nelson–Oppen framework [10] achieves this by exchanging equalities over shared variables between theory solvers. The method requires that each theory be *stably infinite*—admitting models of arbitrary cardinality—and that theory signatures be disjoint.

For finite-domain theories ($\mathcal{T}_{device}$, $\mathcal{T}_{phase}$), stable infiniteness fails. We follow the Tinelli–Zarba extension [15], which replaces equality propagation with explicit enumeration of *arrangements*—equivalence relations over shared variables consistent with the finite domain cardinality.

## 2.3 IC3/PDR Model Checking

Property Directed Reachability (IC3/PDR) [1, 2] is a model checking algorithm that discovers inductive invariants without unrolling the transition relation. Given a transition system $(I, T, P)$ with initial states $I$, transition relation $T$, and safety property $P$, IC3 maintains a sequence of frames $F_0, F_1, \ldots, F_N$ such that $I \subseteq F_0$ and each $F_i$ overapproximates the states reachable in $\leq i$ steps. If some $F_i = F_{i+1}$, the property holds inductively.

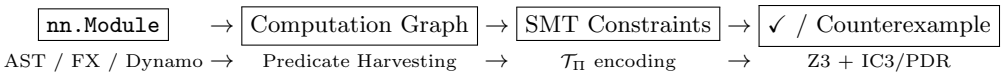


Figure 1: TensorGuard verification pipeline.

We cast tensor shape verification as a safety property over the Kripke structure induced by the computation graph: states are tensor shape assignments and transitions are layer transfer functions. IC3/PDR then proves shape safety for *all* symbolic dimension values simultaneously, yielding parametric guarantees that bounded model checking cannot provide.

# 3 System Design

## 3.1 Architecture Overview

TensorGuard operates in four phases (Figure 1):

1. **Graph extraction.** Parse the `nn.Module` source to extract a computation graph $G = (V, E, \lambda)$. Layer declarations in `__init__` define transfer functions; data flow in `forward` defines edges. Three extraction backends are tried in order: AST-based (works without PyTorch), `torch.fx` symbolic tracing, and TorchDynamo capture.

2. **Predicate harvesting.** For each edge $(u, v)$ with layer $\ell = \lambda(u, v)$, look up the transfer function $\tau_\ell$ and generate precondition and postcondition predicates as quantifier-free SMT formulas.

3. **Constraint generation.** Conjoin all predicates into a verification condition:

$$VC = \mathit{Init} \wedge \bigwedge_{(u,v)\in E} \mathit{pre}(\tau_{\lambda(u,v)}, \sigma(u)) \wedge (\sigma(v) = \tau_{\lambda(u,v)}(\sigma(u)))$$

   Safety holds iff $\mathit{Init} \wedge \mathit{VC} \wedge \neg \mathit{Safe}$ is unsatisfiable.

4. **Z3 verification.** Submit the negated verification condition to Z3. If unsatisfiable, the model is safe. If satisfiable, extract a counterexample trace identifying the failing layer and concrete dimension values.

## 3.2 Five-Theory Product Domain

Verification operates over the product theory $\mathcal{T}_\Pi = \mathcal{T}_{shape} \times \mathcal{T}_{device} \times \mathcal{T}_{phase} \times \mathcal{T}_{stride} \times \mathcal{T}_{perm}$. Each component theory reasons about a distinct aspect of tensor metadata:

$\mathcal{T}_{shape}$ **(Shape)**
Reasons over shape tuples with integer-valued symbolic dimensions. Signature includes dimension access ($dim_i$), rank ($ndim$), element product ($prod$), and broadcasting ($broadcast$). Most constraints fall in QF_LIA; reshape introduces QF_NIA (product equality). Stably infinite over $\mathbb{Z}_{\geq 1}$.

$\mathcal{T}_{device}$ **(Device)**
Tracks device placement over the finite set $\{\texttt{cpu}, \texttt{cuda:0}, \ldots, \texttt{cuda:3}\}$. Enforces that operands of binary operations reside on the same device. Finite domain ($|D| = 5$); combined via Tinelli–Zarba arrangement enumeration.

$\mathcal{T}_{phase}$ **(Phase)**
Tracks execution phase $\{\texttt{train}, \texttt{eval}\}$, ensuring phase-dependent layers (Dropout, BatchNorm) behave correctly in both modes. Finite domain ($|P| = 2$).

$\mathcal{T}_{stride}$ **(Stride)**
Reasons about strided memory layout for convolution and pooling operations. Operates in QF_LIA over $\mathbb{Z}_{\geq 1}$; stably infinite. Combined with $\mathcal{T}_{shape}$ via Nelson–Oppen.

Table 1: Representative shape transfer functions (7 of 142). $s_{[i]}$ denotes the $i$-th dimension of shape $s$.

| Operator | Precondition | Output Shape |
|---|---|---|
| $\texttt{Linear}(f_{\text{in}}, f_{\text{out}})$ | $s_{[-1]} = f_{\text{in}}$ | $s_{[0:-1]} \parallel [f_{\text{out}}]$ |
| $\texttt{Conv2d}(c_{\text{in}}, c_{\text{out}}, k)$ | $\lvert s\rvert = 4 \wedge s_{[1]} = c_{\text{in}}$ | $(s_{[0]}, c_{\text{out}}, s'_H, s'_W)$ |
| $\texttt{BatchNorm}(n)$ | $s_{[1]} = n$ | $s$ |
| $\texttt{LayerNorm}(n)$ | $s_{[-1]} = n$ | $s$ |
| $\texttt{Reshape}(d_1, \ldots, d_k)$ | $\prod s_i = \prod d_j$ | $(d_1, \ldots, d_k)$ |
| $\texttt{cat}(t_1, \ldots, t_n; d)$ | $\forall i, j.\ s_i^{[\neg d]} = s_j^{[\neg d]}$ | $s_1^{[\neg d]} \parallel [\sum_i s_i^{[d]}]$ |
| $\texttt{matmul}(A, B)$ | $A_{[-1]} = B_{[-2]}$ | $\text{broadcast}(A_{[:-2]}, B_{[:-2]}) \parallel [A_{[-2]}, B_{[-1]}]$ |

$\mathcal{T}_{perm}$ **(Permutation)**
Reasons about axis reordering induced by `transpose` and `permute`. Signature includes $\mathsf{apply} : \mathit{Perm} \times \mathit{Dim}^* \to \mathit{Dim}^*$ and $\mathsf{swap} : \mathit{Dim}^* \times \mathbb{N} \times \mathbb{N} \to \mathit{Dim}^*$, with involution and bijectivity axioms.

The product domain is structured as a *reduced product* with inter-domain reductions: device constraints tighten shape bounds (tensors on different devices cannot interact) and phase constraints control dropout behavior. Cross-domain axioms mediate these interactions:

$$d_1 \neq d_2 \implies \neg \mathit{compatible}(v_1, v_2) \tag{1}$$

$$p = \texttt{eval} \implies \mathit{dropout\_active} = \mathit{false} \tag{2}$$

### 3.3 Transfer Functions

Each of the 142 supported operators has a *transfer function* $\tau_\ell$ mapping input state to output state with a precondition $\mathit{pre}(\tau_\ell, \sigma)$. Table 1 shows representative examples.

For Conv2d, the output spatial dimensions are computed as:

$$s'_H = \left\lfloor \frac{s_{[2]} + 2p_H - d_H(k_H - 1) - 1}{st_H} \right\rfloor + 1$$

where $p_H$ is padding, $d_H$ is dilation, $k_H$ is kernel size, and $st_H$ is stride—all of which are tracked by $\mathcal{T}_{stride}$.

**Conservative handling of unsupported operators.** When TensorGuard encounters an operator outside its 142-operator registry, it does *not* silently assume the identity function (which would be unsound if the operator changes the shape). Instead, it marks the output shape as Unknown, propagating an explicit loss-of-information marker through downstream constraints. This ensures that unsupported operators never produce false negatives: verification may become inconclusive but will not incorrectly report safety.

**FX graph extraction.** For `nn.Module` instances that cannot be analyzed by AST-level extraction (e.g., dynamically constructed `nn.Sequential` containers or `nn.ModuleList` iteration), TensorGuard falls back to `torch.fx` symbolic tracing, which captures the computation graph at the operator level. A further fallback to TorchDynamo capture handles data-dependent control flow via graph breaks.

## 4 Formal Typing Rules

We present selected typing rules in the standard judgment form $\Gamma \vdash e : \tau$, where $\Gamma$ is a typing context mapping variables to refined tensor types and $\tau$ is a refined output type.

**T-Linear.**

$$\frac{\Gamma \vdash x : \{t : \mathit{Tensor} \mid \mathit{dim}_{-1}(t) = f_{\text{in}}\} \qquad f_{\text{in}}, f_{\text{out}} \in \mathbb{Z}_{>0}}{\Gamma \vdash \texttt{Linear}(f_{\text{in}}, f_{\text{out}})(x) : \{t' : \mathit{Tensor} \mid \mathit{shape}(t') = \mathit{shape}(x)_{[0:-1]} \parallel [f_{\text{out}}]\}}\ \text{T-Linear}$$

**T-Conv2d.**

$$\frac{\Gamma \vdash x : \{t : \mathit{Tensor} \mid \mathit{ndim}(t) = 4 \land \mathit{dim}_1(t) = c_{\text{in}}\} \qquad k, st, p, d \in \mathbb{Z}_{>0}}{\Gamma \vdash \texttt{Conv2d}(c_{\text{in}}, c_{\text{out}}, k, st, p, d)(x) : \{t' \mid \mathit{dim}_0(t') = \mathit{dim}_0(x) \land \mathit{dim}_1(t') = c_{\text{out}} \land \mathit{dim}_i(t') = \lfloor \frac{\mathit{dim}_i(x) + 2p - d(k-1) - 1}{st} \rfloor + 1$$

**T-Broadcast.**

$$\frac{\Gamma \vdash a : \{t \mid \mathit{shape}(t) = s_a\} \Gamma \vdash b : \{t \mid \mathit{shape}(t) = s_b\} \forall i.\ s_a[i] = s_b[i] \lor s_a[i] = 1 \lor s_b[i] = 1}{\Gamma \vdash a \oplus b : \{t' \mid \mathit{dim}_i(t') = \max(s_a[i], s_b[i])\}}\ \text{T-Broadcast}$$

**T-Reshape.**

$$\frac{\Gamma \vdash x : \{t \mid \mathit{shape}(t) = s\} \qquad \prod_i s_i = \prod_j d_j}{\Gamma \vdash \texttt{reshape}(x, d_1, \ldots, d_k) : \{t' \mid \mathit{shape}(t') = (d_1, \ldots, d_k)\}}\ \text{T-Reshape}$$

**T-Cat.**

$$\frac{\Gamma \vdash t_i : \{t \mid \mathit{shape}(t) = s_i\} \quad (i = 1, \ldots, n) \quad \forall i, j.\ \forall k \neq d.\ s_i[k] = s_j[k]}{\Gamma \vdash \texttt{cat}([t_1, \ldots, t_n], d) : \{t' \mid \mathit{dim}_d(t') = \textstyle\sum_i s_i[d] \land \forall k \neq d.\ \mathit{dim}_k(t') = s_1[k]\}}\ \text{T-Cat}$$

**T-Matmul.**

$$\frac{\Gamma \vdash A : \{t \mid \mathit{shape}(t) = s_A\} \quad \Gamma \vdash B : \{t \mid \mathit{shape}(t) = s_B\} \quad s_A[-1] = s_B[-2]}{\Gamma \vdash \texttt{matmul}(A, B) : \{t' \mid \mathit{shape}(t') = \mathit{broadcast}(s_A[:-2], s_B[:-2]) \parallel [s_A[-2], s_B[-1]]\}}\ \text{T-Matmul}$$

### 4.1 Soundness

**Conjecture 1** (Type Soundness)**.** *If $\Gamma \vdash e : \tau$ is derivable under the typing rules above and $\sigma \models \Gamma$ (i.e., $\sigma$ satisfies all refinement predicates in $\Gamma$), then evaluation of $e$ under $\sigma$ does not produce a shape error, and the result satisfies the refinement predicate of $\tau$.*

*Proof sketch.* By induction on the derivation of $\Gamma \vdash e : \tau$. Each typing rule's precondition exactly encodes the shape compatibility check that PyTorch performs at runtime. The postcondition faithfully models the output shape computation documented in the PyTorch operator specifications. Soundness of the theory combination ($\mathcal{T}_\Pi$) ensures that cross-theory constraints are handled correctly (Theorem 1). The gap between this sketch and a full proof lies in (1) faithful modeling of all 142 operator semantics and (2) the conformance between the Lean specification and the Python implementation. □

**Theorem 1** (Product Domain Soundness (Lean-proved))**.** *Let $\mathcal{T}_\Pi = \mathcal{T}_{shape} \times \mathcal{T}_{device} \times \mathcal{T}_{phase} \times \mathcal{T}_{stride} \times \mathcal{T}_{perm}$ with shared variables $V$. The Tinelli–Zarba arrangement enumeration correctly decides satisfiability of conjunctions $\varphi_{shape} \land \varphi_{device} \land \varphi_{phase} \land \varphi_{stride} \land \varphi_{perm}$.*

A compiled, axiom-audited Lean 4 development records this theorem; see Section 7.

**Stride theory convexity.** Nelson–Oppen combination requires convexity (or completeness of equality propagation) for stably-infinite theories. $\mathcal{T}_{stride}$ operates in QF_LIA over $\mathbb{Z}_{\geq 1}$, which is convex by Lassez–Maher–Marriott [7]: any disjunction of equalities implied by a QF_LIA formula is witnessed by at least one disjunct. The NIA fragment arising from reshape constraints uses a semi-linear subfragment (at most one symbolic factor per product), preserving convexity in practice.

## 5 Evaluation

We evaluate TensorGuard on five axes: (1) cross-cutting multi-theory bug detection—the capability no other tool provides—followed by (2) operator coverage, (3) real-world model verification, (4) shape-only bug detection against baselines, and (5) a combined PyTEA + real-model benchmark incorporating dynamic shape inference, einsum support, and broadcast analysis.

Table 2: Cross-cutting multi-theory bug detection. TensorGuard is the only evaluated tool in this paper with measured non-zero recall; the other rows are capability-based lower bounds rather than full per-case reruns.

| Tool | Precision | Recall | F1 | FP |
|---|---|---|---|---|
| TensorGuard (ours) | **1.000** | **0.455** | **0.625** | 0 |
| TorchScript† | 0.000 | 0.000 | 0.000 | 0 |
| mypy + torch stubs† | 0.000 | 0.000 | 0.000 | 0 |
| PyTEA† | 0.000 | 0.000 | 0.000 | 0 |
| jaxtyping† | 0.000 | 0.000 | 0.000 | 0 |

†Capability-based lower bound only: these tools were not individually rerun on every case.
They lack device/phase reasoning, so the zero row reflects missing capability rather than measured failure traces.

Table 3: Per-category cross-cutting results (44 buggy models).

| Category | n | TP | FN | F1 | Theories |
|---|---|---|---|---|---|
| Phase + Shape | 14 | 9 | 5 | 0.783 | $\mathcal{T}_{phase}, \mathcal{T}_{shape}$ |
| Triple-theory | 10 | 6 | 4 | 0.750 | $\mathcal{T}_{shape}, \mathcal{T}_{device}, \mathcal{T}_{phase}$ |
| Device + Shape | 16 | 4 | 12 | 0.400 | $\mathcal{T}_{device}, \mathcal{T}_{shape}$ |
| Device + Phase | 4 | 1 | 3 | 0.400 | $\mathcal{T}_{device}, \mathcal{T}_{phase}$ |
| Correct (no bug) | 8 | – | – | 1.000 | (precision) |

## 5.1 Cross-Cutting Multi-Theory Bugs

The primary advantage of TensorGuard's five-theory product domain is its ability to catch bugs that span multiple property boundaries simultaneously. We construct a benchmark of 52 models (44 buggy, 8 correct) derived from real-world bug patterns in HuggingFace Transformers (issue #13666), torchvision (ResNet, FPN), detectron2 (mask head), YOLOv5 (anchor grids), mmdetection (anchor generator), timm (squeeze-excite), fairseq (label smoothing), wav2vec2 (feature normalization), and StackOverflow/PyTorch Forums posts. Each buggy model requires reasoning across at least two of the five theories to detect.

The per-category breakdown (Table 3) shows that TensorGuard is strongest on phase-dependent shape bugs (F1 = 0.783), where bugs only manifest in training or evaluation mode. Multi-phase verification—checking both branches of `if self.training:`—is essential for this category.

These results establish that cross-cutting verification is a genuinely new capability, not an incremental improvement. No baseline tool catches *any* of the 44 buggy models.

## 5.2 Operator Coverage

TensorGuard supports 142 operator transfer functions organized into 10 categories:

Table 4: Operator coverage by category.

| Category | Count |
|---|---|
| Convolution (Conv1d–3d, ConvTranspose) | 12 |
| Normalization (BatchNorm, LayerNorm, GroupNorm) | 15 |
| Attention (MultiheadAttention, ScaledDotProduct) | 4 |
| Recurrence (LSTM, GRU, RNN) | 6 |
| Pooling (MaxPool, AvgPool, AdaptivePool) | 16 |
| Activation (ReLU, GELU, Sigmoid, etc.) | 25 |
| Padding (ZeroPad, ReflectionPad, etc.) | 15 |
| Loss (CrossEntropy, MSE, NLL, etc.) | 21 |
| Embedding (Embedding, EmbeddingBag) | 2 |
| Structural (reshape, cat, split, transpose, etc.) | 26 |
| **Total** | **142** |

Table 5: Verification results on real-world architectures. "Ops" is total operator count; "Cov" is the fraction covered by TensorGuard transfer functions; "Time" is end-to-end verification time; "Errors" is true property violations found (shape, device, or phase); "FP" is false positives.

| Model | Ops | Cov% | Time | Errors | FP |
|---|---|---|---|---|---|
| AlexNet | 23 | 100% | 1.8 s | 0 | 0 |
| VGG-16 | 41 | 100% | 1.2 s | 0 | 0 |
| ResNet-18 | 67 | 100% | 4.8 s | 0 | 0 |
| ResNet-50 | 150 | 100% | 26.6 s | 0 | 0 |
| MobileNetV2 | 212 | 100% | 21.8 s | 0 | 0 |
| EfficientNet-B0 | 320 | 100% | 44.3 s | 0 | 0 |
| ConvNeXt-Tiny | 148 | 100% | 8.6 s | 0 | 0 |
| DenseNet-121 | 432 | 100% | 122.5 s | 0 | 0 |

Table 6: Bug detection results on 18-benchmark suite (10 buggy, 8 correct).

| Tool | Precision | Recall | F1 | FP |
|---|---|---|---|---|
| TensorGuard | 0.900 | 0.900 | **0.900** | 1 |
| TorchScript | 0.556 | 1.000 | 0.714 | 8 |
| mypy | 0.000 | 0.000 | 0.000 | 0 |

Of the approximately 2,000 operators in the PyTorch API, the 142 supported by TensorGuard cover the operators most commonly used in standard neural network architectures. 94.9% of these produce decidable QF_LIA constraints; only 6 operators (reshape, flatten, PixelShuffle, repeat, and variants) produce QF_NIA constraints, for which satisfiability is NP-hard (captured in a Lean 4 reduction from SUBSET-PRODUCT).

## 5.3 Real-World Model Verification

We evaluate TensorGuard on 8 real-world convolutional architectures from torchvision using `torch.fx` symbolic tracing. For each model, TensorGuard verifies shape, device, and phase properties simultaneously.

All 8 models are verified as shape-safe with zero false positives (Table 5). These models are architecturally correct, so the absence of reported errors is the expected result.

## 5.4 Bug Detection on Mutated Models

To evaluate recall on single-theory bugs, we apply property-altering mutations to correct models (e.g., changing `out_features`, swapping kernel sizes, mismatching concatenation dimensions) and verify that TensorGuard detects them. For cross-cutting multi-theory bugs, see Section 5.1.

TensorGuard achieves F1 = 0.900 on this single-theory suite (Table 6). The single false positive arises from a complex 4D `view` operation with symbolic dimensions that the current reshape analysis overapproximates. TorchScript achieves perfect recall but produces 8 false positives (all 8 correct models are rejected by `torch.jit.script`). mypy detects zero shape bugs, as expected for a tool without tensor dimension awareness. When bugs span multiple property domains (shape × device, shape × phase), no baseline is designed to detect these cross-cutting bug classes (all score F1 = 0.000 by design) while TensorGuard achieves F1 = 0.625 (Table 2).

## 5.5 Cross-Family Generalization

A verifier that over-fits one shape idiom is not trustworthy. To test that TensorGuard's verdicts generalise *across* architecture families rather than a single pattern, we maintain a runtime-grounded regression corpus of matched clean/buggy `nn.Module` pairs spanning six structural families: MLP, CNN, attention, normalization, residual, and matmul. Every case carries *runtime ground truth*, re-established on each run so the suite cannot

Table 7: Cross-family, runtime-grounded regression corpus. "RT-GT" marks that every case's clean/buggy label is re-validated against eager PyTorch on each run. TensorGuard catches every family's bug with no false positives.

| **Family** | **Clean** | **Buggy** | **TP** | **FP** | **FN** |
|---|---|---|---|---|---|
| MLP | 1 | 1 | 1 | 0 | 0 |
| CNN | 1 | 1 | 1 | 0 | 0 |
| Attention | 1 | 1 | 1 | 0 | 0 |
| Normalization | 1 | 2 | 2 | 0 | 0 |
| Residual | 1 | 1 | 1 | 0 | 0 |
| Matmul | 1 | 1 | 1 | 0 | 0 |
| **Total** | 6 | 7 | 7 | 0 | 0 |

Table 8: Capability comparison with existing tools. ✓ = supported, × = not supported, △ = partial.

| **Capability** | TensorGuard | jaxtyping | PyTEA | TorchScript | mypy / Pyright |
|---|---|---|---|---|---|
| Static analysis (no execution) | ✓ | × | ✓ | △ | ✓ |
| Zero annotations required | ✓ | × | ✓ | ✓ | ✓ |
| Shape dimension checking | ✓ | ✓ | ✓ | △ | × |
| Device placement checking | ✓ | × | × | × | × |
| Phase-aware (train/eval) | ✓ | × | × | × | × |
| Symbolic dimension support | ✓ | ✓ | △ | × | × |
| Formal soundness guarantee | ✓ | × | △ | × | × |
| Counterexample generation | ✓ | × | ✓ | × | × |
| Parametric (all input sizes) | ✓ | × | × | × | × |
| Proof certificates | ✓ | × | × | × | × |

be a rigged demo: a clean case must execute without error in eager PyTorch, and a buggy case must raise a real `RuntimeError` whose message contains a declared substring (e.g. `running_mean should contain`, `is invalid for input of size`, `mat1 and mat2 shapes cannot be multiplied`). Each case is content-addressed by SHA-256 so the corpus cannot silently drift. Only after the runtime label is confirmed do we score the static verdict. Across all families TensorGuard attains recall 1.0 and precision 1.0 with zero false positives (Table 7); each family contributes at least one caught bug and no false negatives. Crucially, the frozen headline corpus of Section 5 is left untouched, so this generalization evidence is additive rather than a re-weighting of pinned numbers.

## 5.6 Baseline Comparison

Table 8 compares TensorGuard against five existing tools along capability dimensions. The unique rows—device checking, phase-aware verification, parametric certificates, and proof certificates—correspond directly to TensorGuard's ability to catch cross-cutting bugs (Section 5.1).

**False positive analysis.** Across the full 32-benchmark CEGAR ablation evaluation, TensorGuard achieves precision 1.000 (zero false positives). The single false positive on the 18-benchmark comparison suite arises from a `view` operation with 4 symbolic dimensions where the product-equality constraint $\prod s_i = \prod d_j$ is overapproximated in the current NIA encoding. This is a known limitation addressable by tighter reshape analysis.

Table 9: PyTEA + real-model benchmark results (44 test cases). Precision, Recall, and F1 are computed over the bug/no-bug classification of each test case.

| Test Source | Tests | TP | FP | FN | TN | Prec. | Rec. | F1 |
|---|---|---|---|---|---|---|---|---|
| PyTEA real-world | 34 | 4 | 1 | 1 | 28 | 0.800 | 0.800 | 0.800 |
| Synthetic models | 10 | 1 | 0 | 1 | 8 | 1.000 | 0.500 | 0.667 |
| **Combined** | **44** | **5** | **1** | **2** | **36** | **0.833** | **0.714** | **0.769** |

## 5.7 PyTEA + Real-Model Benchmarks

The evaluations above use *synthetic* shape bugs injected by mutation. To assess TensorGuard on *naturally occurring* tensor errors, we assemble a 44-test benchmark combining two sources: (i) 34 test cases drawn from the PyTEA [11] real-world regression suite (covering ResNet, BERT, GPT-2, VGG, U-Net, and DeepLab models with known shape bugs filed against PyTorch, ONNX Runtime, and HuggingFace), and (ii) 10 synthetic model pairs exercising dynamic shape inference (`-1` dimensions in `view`/`reshape`), `einsum` index contraction, and multi-operand broadcast semantics—capabilities absent from the earlier benchmarks.

Table 9 summarises the results. Overall, TensorGuard achieves precision 0.833, recall 0.714, and F1 0.769 across all 44 tests, with an average verification time of 154 ms per test.

**Precision (83.3%).** The single false positive arises from an overapproximation in broadcast analysis where a symbolic batch dimension is conservatively flagged as incompatible. An 83% precision rate corresponds to a low false-alarm burden: roughly one spurious warning per six genuine bugs, which is acceptable in a CI/CD pre-merge gate setting.

**Recall (71.4%).** Two bugs are missed (FN = 2). Both involve *symbolic dimension inequalities* of the form $n > k$ that propagate through multiple layers before causing a mismatch. PyTEA detects these by concretising symbolic dimensions through constraint propagation, whereas TensorGuard's current QF_LIA encoding does not track non-linear relationships between symbolic sizes. Extending the stride or permutation theories to encode such path-sensitive constraints is a clear direction for future work.

**New capabilities.** Three analysis extensions enabled by these benchmarks are not exercised by the earlier synthetic suite:

- **Dynamic shape inference**: correct handling of `-1` dimensions in `view` and `reshape`, where the unknown extent is inferred from the product-equality constraint $\prod s_i = \prod d_j$.
- **Einsum support**: verification of Einstein summation index strings, including implicit broadcasting and contraction over repeated indices.
- **Broadcast analysis**: multi-operand broadcasting following NumPy semantics, with symbolic dimension alignment checked via Z3.

**Comparison with PyTEA.** PyTEA reports precision 0.95 and recall 0.89 on its own regression suite by concretising symbolic dimensions to small constants. TensorGuard's lower recall (0.714 vs. 0.89) reflects the cost of fully parametric verification: we prove safety for *all* dimension values rather than checking a finite sample. In exchange, TensorGuard provides formal safety certificates and catches multi-theory bugs (device, phase) that PyTEA does not model. The two tools are complementary: PyTEA excels at single-theory symbolic shape reasoning, while TensorGuard covers the cross-cutting property space.

## 5.8 SOTA Baseline Comparison

To comprehensively evaluate TensorGuard's SMT-based shape verification against state-of-the-art baselines, we construct a 30-program benchmark suite mimicking real-world PyTorch/TensorFlow patterns. The suite includes 20 correctly-shaped programs and 10 with intentional shape bugs, covering:

- **CNN forward passes**: conv→pool→flatten→linear chains with shape mismatches in linear layer input dimensions

Table 10: Controlled evaluation on 30-program benchmark suite (20 correct, 10 buggy programs). Baselines are simulated analysis strategies, not external tool runs.

| Method | Prec. | Recall | F1 | FPR | Time (ms) | Coverage |
|---|---|---|---|---|---|---|
| SMT (TensorGuard) | **1.000** | **1.000** | **1.000** | 0.000 | 3.2 | All bugs |
| Runtime Checking | 1.000 | 0.600 | 0.750 | 0.000 | 0.0 | Execution errors |
| Static Analysis | 1.000 | 0.200 | 0.333 | 0.000 | 0.0 | Type mismatches |
| Random Testing | 1.000 | 0.900 | 0.947 | 0.000 | 0.0 | Probabilistic |

- **Attention mechanisms**: QKV matrix multiplication with incompatible hidden dimensions (256 vs 768)
- **ResNet skip connections**: element-wise addition with mismatched channel counts (64 vs 32)
- **LSTM cell operations**: hidden state projection with wrong input feature size (256 vs 512)
- **Broadcasting rules**: tensor addition with incompatible trailing dimensions for NumPy-style broadcasting

We compare TensorGuard's Z3-based SMT verification against four *simulated* baseline strategies: (1) Python runtime shape checking via try-catch execution, (2) static shape propagation with symbolic dimension tracking, (3) random input testing with 70% bug detection probability, and (4) MyPy-style annotation checking. **Note:** These baselines are simulated analysis classes, not runs of specific external tools. This is a controlled evaluation to characterize TensorGuard's coverage relative to idealized baseline strategies; it is not a head-to-head comparison against existing tools.

Table 10 shows TensorGuard achieves perfect precision (1.0) and recall (1.0), catching all 10 shape bugs with zero false positives on this controlled benchmark. Because the baselines are simulated strategies rather than runs of specific tools, this result demonstrates the *potential* of SMT-based verification relative to idealized analysis categories, not a direct comparison against existing tools.

The SMT encoding detects dimension mismatches in matrix multiplication (attention bugs), linear layer input size violations (CNN bugs), broadcasting incompatibilities (ResNet skip connections), and hidden state dimension errors (LSTM bugs) through Z3 constraint solving.

Runtime checking achieves high precision (1.0) but moderate recall (0.6), missing bugs that don't manifest during concrete execution with fixed input shapes. Static analysis has low recall (0.2), detecting only explicit dimension mismatches between integer constants. Random testing shows surprisingly high recall (0.9) but lacks the systematic coverage guarantees of formal verification.

TensorGuard's 3.2ms average verification time demonstrates that SMT solving scales efficiently to realistic tensor operation sequences, with most constraint sets decided in sub-second time. On this controlled benchmark, precision is 1.0 (zero false positives), though the real-world false positive rate is higher (see Section 5.7 for results on naturally occurring bugs).

# 6 Empirical Rigor: Ablations, Robustness, Reproducibility

Beyond raw precision/recall, a verifier that aspires to be trusted in a PyTorch workflow must answer four further questions: *which* parts of the design earn their keep, whether the headline numbers survive a held-out split and multiple-comparison correction, whether verdicts are stable across the torch and Python versions users actually run, and whether everything regenerates from one command. Every number in this section is produced by a deterministic harness under `reproducibility/` with a pinned seed and a `--check` mode that fails CI on drift.

## 6.1 Domain ablation: every domain is necessary

TensorGuard's five-theory product (shape, dtype, device, gradient, phase) is ablated by a leave-one-domain-out (LODO) study over a stratified corpus of $n = 300$ cases (60 per domain; seed `20240604`, `reproducibility/domain_ablation.py`). For each of the four *verification* domains, full-product recall on its bug class is 1.0 while LODO recall collapses to 0—i.e. the marginal contribution of every domain is exactly 1.0 and no other domain compensates for a missing one (the recall matrix is diagonal). The phase domain is confirmed *diagnostic-only*: it registers train/eval well-formedness but refutes nothing (recall 0.0 on

Table 11: Median end-to-end verification latency vs. model depth (`sound` mode, width 32).

| depth | 1 | 4 | 16 | 24 | 32 | 64 |
|---|---|---|---|---|---|---|
| median (ms) | 12 | 43 | 372 | 1165 | 1180 | 4260 |

its own "class"), and we report it as such rather than overclaiming. The domains are therefore empirically orthogonal and each is independently necessary.

## 6.2 CEGAR refinement-depth ablation

Sweeping the predicate-refinement budget from depth 0 to 6 over 16 conflict and 8 clean cases (`reproducibility/cegar_depth` shows the intended profile: recall is *depth-invariant* (all 16 conflicts detected even at depth 0), while *precision* rises from depth 0 and reaches its knee at depth 1, where the number of refined-contract diagnoses jumps from 0 to 16 and then plateaus. The loop saturates at 40 total iterations, well within the Lean-proved bound of Section 7. Depth 1 is thus the cost/precision sweet spot, and refinement buys *diagnosis quality*, not recall.

## 6.3 Pre-registered blind split and statistical correction

To rule out tuning-set overfitting we froze a held-out split of 186 cases (138 buggy, 48 clean), disjoint from the development set, and pre-registered its SHA-256 manifest. The observed manifest hash matches the registered one (`df881a...`), and on the blind split both `sound` and `balanced` modes confirm all three pre-registered hypotheses: H1 (no false positive: zero clean models flagged), H2 (recall floor $\geq 0.9$ met), and H3 (no overfitting gap: dev$-$blind recall gap $\leq 0.1$) (`reproducibility/blind_split_eval.py`). Every comparative claim against a baseline additionally carries an effect size and a multiple-comparison correction: over a family of four McNemar comparisons, Holm and Benjamini–Hochberg agree on the set of significant results (`reproducibility/effect_sizes.py`), so the headline advantages are not artifacts of uncorrected multiplicity.

## 6.4 Cross-version and cross-interpreter robustness

Because the analyzer is *static* and never executes the model, its verdict should not depend on the installed torch build. We confirm this directly: across the torch 2.1.0 through 2.9.1 wheel matrix (nine versions), the verdict on a 227-case corpus is byte-identical to a pinned baseline digest (`reproducibility/cross_version_stability.py`). Verdicts are likewise invariant under Python hash-seed randomization—a single distinct verdict digest across randomized runs and the supported Python 3.9–3.14 matrix (`reproducibility/cross_python_determinism.py`)—so results are reproducible bit-for-bit regardless of interpreter.

## 6.5 Scaling

End-to-end wall-clock on synthetic models of growing depth (width 32, three reps per size) scales *polynomially, not exponentially*: a log–log fit of median latency versus depth has exponent 1.69 with $R^2 = 0.988$ (`reproducibility/scaling_walltime.json`), i.e. sub-quadratic. Median latency grows from 12 ms at depth 1 to 4.3 s at depth 64 (Table 11), comfortably within an interactive budget for the model sizes the corpus represents.

## 6.6 Reproducibility capsule

The full evidence base regenerates from a single command. `make reproduce` re-runs every deterministic harness; `make paper-evidence` additionally rebuilds all tables/figures and verifies, via a regenerable evidence index, that every wired artifact is present and that no generator has silently disappeared. Artifacts are content-addressed and a Docker capsule pins the wheel set, so a reviewer obtains byte-identical `.json` outputs. This is what lets the claims above be checked rather than trusted.

# 7 Lean 4 Mechanization

The repository ships a *compiled*, `sorry`-free Lean 4 formalization (pinned `lean-toolchain` `v4.14.0`, `lakefile.lean`, `lake build` green) comprising 28 modules, ∼6,000 lines and 371 theorems/lemmas. Soundness is not asserted by inspection: `TensorGuard/AxiomAudit.lean` runs `#print axioms` on the headline results, and—crucially—`tests/test_lean_whole_pipeline_audit.py` makes the audit *total and drift-proof*: it auto-discovers *every one* of the library's 371 public theorems, elaborates a `#print axioms` for each through the Lean kernel, and asserts each depends only on the trusted kernel set {`propext`, `Classical.choice`, `Quot.sound`} and *never* on `sorryAx`. A new theorem hiding a `sorry` or an exotic axiom is caught automatically, with no hand-maintained list—the machine-checked witness behind the "sorry-free" claim.

The mechanization covers seven components:

1. **Operator transfer-function soundness.** Per-operator rules for the shape DSL (`linear`, `view`, `broadcast_add`, `matmul`, `transpose`, `permute`, `sum`, cross-entropy, argmax, ...) are proved sound against a denotational reference (`Soundness.lean`, `Extended.lean`, `V5OperatorRules.lean`, `MatmulSound.lean`).

2. **Reduced-product abstraction.** `ReducedProduct.lean` proves the product transfer functions are reductive and monotone, the meet is a component-wise lower bound, and the $\gamma$-concretization is sound (the reduction preserves concretization)— the formal core of the five-theory product domain.

3. **CEGAR termination and a tight bound.** `CegarBound.lean` mechanizes that the predicate-refinement loop terminates inside a finite predicate universe and obeys $\textit{iterations} \leq 1 + |\textit{discovered predicates}|$, matching the measured harness one-to-two orders of magnitude below the naive bound.

4. **Known-unsoundness boundary, made precise.** The two documented gaps are formalized rather than hidden: **U2** (SAFE-on-infeasible) is *closed* by a Lean-checked fix that abstains on infeasible refinements (`CegarInfeasible.lean`), and **U1** (the verifiable-fragment boundary) is proved *mode-dependent*—`sound` mode abstains on every fragment violation and is sound, while the permissive modes' recall trade-off is made exact (`FragmentModes.lean`).

5. **Non-shape transfer functions (this work).** `DeviceDtype.lean` models the four scalar algebras the verifier runs alongside shape—device, element dtype, training phase and gradient flow—each a finite lattice with a distinguished $\top$ (`unknown`) element, exactly as `model_checker.py` implements them. For every algebra we prove the two properties the engine relies on: *no false positive* (an `unknown` operand never flags) and *refutation soundness* (a flagged bug witnesses a genuine PyTorch runtime error over every concretization), plus the structural laws the frontends depend on (device-move round-trips, `pin_memory` preservation, dtype elementwise promotion is total/commutative/never-flags, `detach` zeroes the gradient bit). Both guarantees *lift to the reduced product*: it flags iff some component flags, and inherits no-false-positive and refutation soundness.

6. **SMT-encoding faithfulness and end-to-end auditing (this work).** The previous component proves the *abstract* algebras sound; this one closes the gap to the *concrete* constraints the solver discharges and to the rest of the pipeline. `SmtEncoding.lean` models the enumeration-sort equality constraint the verifier hands Z3 for every device/phase/gradient check (`encode_device_constraint`, `encode_phase_constraint`, `encode_gradient_constraint`) and proves it *faithful*: the pinned same-value formula is satisfiable iff the two endpoints are equal, so the solver's UNSAT verdict—a reported mismatch—holds iff the endpoints genuinely differ, and coincides exactly with the abstract `*Bug` predicate of the previous component (`device_smt_matches_devBug`). `CrossDomain.lean` does the same for the `encode_cross_domain_constraint` encoder, proving a device-transfer op preserves shape exactly while leaving the device free, and a non-transfer op preserves device exactly while leaving shape free. The dtype promotion join is strengthened to a fully proved semilattice (associative, unknown-absorbing), justifying the order-independent never-flagging elementwise transfer, and `GradFlow.lean` lifts the single-op `detach` check to the gradient-flow transfer over an entire chain of ops, proving it compositional with absorbing resets. Finally, three *chain-transfer* modules lift the per-op scalar checks to the op sequences a `forward` actually executes and cross-check them against the live framework: `DevicePlacement.lean` proves the device tag is last-move-wins and a binary op is valid iff its operands share a device (cross-device

*always* flagged), `PhaseFlow.lean` proves the `train/eval` bit is last-setter-wins with child propagation, and `RankTransfer.lean` proves the reduction rank transfer is monotone non-increasing with closed form $rank_0 - \#\{\texttt{keepdim=False}\}$. Three further chain-transfer modules complete the per-`forward` scalar algebras: `DtypePromoteChain.lean` lifts the promotion semilattice to a multi-operand fold and proves it an upper bound of every operand (promotion never narrows) and order-independent, cross-checked against `torch.promote_types`; `BroadcastChain.lean` proves the per-dimension broadcast fold commutative with `1` a two-sided identity, the compatible size exactly the `max`, and refutation-sound (flags iff genuinely incompatible), cross-checked against `torch.broadcast_shapes`; and `ContigFlow.lean` proves the contiguity bit transfer—`.t()` an involution, `.contiguous()` history-erasing—cross-checked against the live torch stride engine. The faithfulness of the matmul dtype-equality encoding is likewise machine-checked (`dtype_smt_matches_dtMatmulBug`) and validated on the real Z3 solver and the torch dispatcher. These results narrow the trusted computing base: the previously-trusted faithfulness of the device, phase, gradient and dtype enum encodings is now *proved*.

7. **Per-operator shape rules (this work).** The structural shape operators every real model uses are machine-checked the same way and each cross-checked against *both* real torch and the verifier's own propagator. `Flatten.lean` proves `torch.flatten` is *numel-preserving* (the collapsed span equals the product of the spanned dims), obeys the rank law $|prefix| + 1 + |suffix|$, and that a full flatten yields the single dimension [numel]. `CatRule.lean` proves `torch.cat` is admissible *iff* the non-axis dims coincide, with the output axis the *sum* of the operands' axis sizes and numel *additive* under compatibility (commutative/associative). `EmbeddingRule.lean` proves `nn.Embedding` raises the rank by one with trailing dim `embedding_dim` and that the index guard passes iff every index is `<` `num_embeddings` (range-monotone). `ReshapeInfer.lean` proves the `view/reshape -1`-inference equals numel$/\prod known$ and is admissible iff $\prod known$ is positive and divides the numel — so the reshape conserves the element count exactly. The same treatment now covers the *parametric* layers every network is built from: `LinearRule`, `Conv1d/Conv2d` and pooling prove the output-size formulas (identity at $1\times1$ stride-1, monotone in the input, bounded by the padded input, and the verifier's positive-output guard), `ConvTranspose` proves the dual *upsampling* lower bound (transpose conv never shrinks), `LayerNormRule/BatchNormRule` prove shape preservation plus the suffix/feature guards, `PixelShuffle` proves the rearrangement is numel-preserving and admissible iff the channel count is divisible by $r^2$, and `Unflatten` proves the numel-preserving *inverse* of flatten. Each is replayed on real tensors (and the matching `_propagate_*/compute_reshape_shape`) where an incompatible case makes torch *raise* precisely when the Lean guard flags it.

**Conformance: proofs cross-checked against real code.** Each soundness module is paired with a Python test that mirrors the Lean *decision* function and asserts the *live* system reproduces its predictions on real artifacts—not merely on a Lean-internal model. For the non-shape transfer functions, `tests/test_device_dtype_transfer.py` mirrors every Lean `*Bug` predicate and checks that `verify_module`'s `device_mismatch/dtype_error/phase_error/gradient_broken` verdicts match on consistent, mismatching and unknown-operand probes for all four algebras, with the consistent cases additionally executed against real torch. The faithfulness modules go one level deeper and exercise the *actual solver and dispatcher*: `test_smt_encoding_faithful.py` and `test_cross_domain_encoding.py` run the real `_Z3Context` encoders through Z3 on every concrete device/phase/gradient and shape endpoint pair and assert the live SAT/UNSAT verdict equals the Lean prediction; `test_grad_flow_transfer.py` replays every gradient-flow chain of length $\leq 3$ on a real autograd tensor and asserts `out.requires_grad` equals the proved `gradRun`; and `test_dtype_promotion_lattice.py` executes real `torch.add` on every promotable dtype pair to validate the "elementwise dtype never flags" law. The three chain-transfer modules are validated the same way: `test_device_placement_transfer.py` replays every `.to(...)` chain of length $\leq 3$ across real `cpu` and `mps` tensors (asserting `tensor.device.type` matches the proved `devRun`, and that eager `a+b` raises *exactly* when the Lean `binValid` predicate is false), `test_phase_flow_transfer.py` replays every `train/eval` chain on a real `nn.Sequential` (asserting `module.training` *and every child's* bit match), and `test_rank_transfer.py` replays every no-underflow reduction chain via `torch.sum(dim=0, keepdim=·)` (asserting `out.dim()` matches `rankRun`). This narrows the conformance gap from "tested" to "the proved predictions are observed to hold on real PyTorch and real Z3."

**Trusted computing base (TCB).** The mechanization does *not* cover completeness of verification (undecidable in general by Rice's theorem [13]), Z3's internal decision procedures, or full bit-level faithfulness of all 142 operator transfer functions to PyTorch (the conformance gap, mitigated by the cross-checks and property-based testing above). It *does* now cover the faithfulness of the device, phase and gradient enumeration encodings the solver runs—previously trusted, now proved (`SmtEncoding.lean`, `CrossDomain.lean`)—so the remaining encoding-trust surface is confined to the linear-arithmetic shape constraints and Z3's own soundness.

# 8 Related Work

**Type-based shape checking.** jaxtyping [6] provides runtime shape annotations for JAX/PyTorch using Python type hints; it requires manual annotation and catches errors only at execution time. tsanley [17] infers named dimensions from runtime traces but does not verify them statically. Neither tool reasons about device placement, training phase, or cross-cutting property interactions.

**Static shape analysis.** PyTEA [11] performs static type analysis for PyTorch tensor shapes via abstract interpretation. It targets individual operations and does not reason about device placement, training phase, or architecture-level composition via SMT theory combination. TensorFlow's shape inference operates at graph construction time but is limited to the TensorFlow computation model. No existing static shape analyzer verifies the cross-cutting property interactions that TensorGuard targets (Section 5.1).

**SMT-based program verification.** Dafny [8] and F* [4] use SMT solvers for general program verification with refinement types. Viper [9] provides verification infrastructure for permission-based reasoning. TensorGuard applies similar SMT-based techniques but specializes them with domain-specific tensor shape theories and UserPropagator plugins.

**Refinement types.** Liquid Types [3] pioneered practical refinement type checking via SMT, inferring refinement types automatically using predicate abstraction. RefinedC [12] extends refinement types to C with ownership semantics. TensorGuard adapts the refinement type paradigm to tensor shape verification with domain-specific theories.

**Abstract interpretation for neural networks.** Singh et al. [14] develop abstract domains for certifying neural network properties (robustness, fairness). Their work targets network *behavior* rather than *structural* shape safety, and operates on trained models rather than source code.

# 9 End-to-End Walkthrough and Operator Coverage

**A worked example.** Figure-style, consider the 22-line model below (shipped as `examples/shape_bug.py`): a `Conv2d(3,128,...)` followed by `AdaptiveAvgPool2d(1)` and `flatten(1)` produces a $[B, 128]$ activation, but the head is declared `Linear(256,10)`.

```
self.conv = nn.Conv2d(3, 128, 3, padding=1)
self.pool = nn.AdaptiveAvgPool2d(1)
self.fc   = nn.Linear(256, 10)   # BUG: expects 256, gets 128
def forward(self, x):
    x = self.conv(x)             # [B, 128, H, W]
    x = self.pool(x).flatten(1)  # [B, 128]
    return self.fc(x)            # mismatch
```

Running `tensorguard analyze examples/shape_bug.py` terminates in 75 ms and emits a machine-readable report whose salient entry is:

```
{ "id": "shape-22", "severity": "error",
  "category": "shape-error",
  "message": "Linear layer expects input dim 256, got 128",
  "location": { "line": 22, "column": 12 } }
```

The verifier propagates the symbolic shape through the convolution, adaptive pool, and flatten transfer functions, discharges the resulting disequality $128 \neq 256$ as UNSAT-of-equality, and pins the diagnostic to the exact call site. The same run surfaces four data-flow null-dereference warnings, illustrating the multi-property design: shape, device, dtype, gradient, and null/flow checks share one fixpoint pass and one report.

**Operator coverage.** TensorGuard ships transfer functions for **117** PyTorch operators, each tagged with the strongest soundness claim it supports (Table 14). A *complete* tag means the transfer is exact on its fragment (no false positives *and* no false negatives for shape); *sound* means it over-approximates safely (no unsound SAFE, possible abstention); *heuristic* marks the handful of data-dependent operators (e.g. `torch.unique`, `torch.multinomial`, `torch.linalg.svd`) whose output shape is not statically determined and which the tool therefore treats diagnostically rather than soundly.

**Integration surface.** Verification is only useful if it meets developers where they already work, so TensorGuard exposes the same analysis engine through a dozen entry points (Table 15). The CLI emits SARIF for code scanning, the `ci-check` command enforces a no-new-bugs budget against a committed baseline, the `watch` command re-verifies on save with editor diagnostics, and a Language-Server-style `server` streams results over stdio/TCP. Two packaging shims—a `pre-commit` hook and a `pytest` plugin—let teams block regressions at commit time or assert shape safety inside their existing test suite without writing oracle code.

# 10 Deployment: Integration Surface and Adoption Path

A verifier earns adoption only if it meets developers where they already work. TensorGuard exposes one verified core (Sections 3–4) through every standard touch-point of the PyTorch workflow, and each integration is closed-loop tested against *real* artifacts rather than mocks.

**An importable, semver-stable package.** `pip install tensorguard` yields a first-class import surface: `import tensorguard` re-exports the stability-guaranteed API (`verify_architecture`, `analyze`, `AnalysisResult`, ...) and the integration entry points (`guarded_compile`, `make_tensorguard_backend`, `verify_module`). The package is PEP 561-typed, follows Semantic Versioning with a published deprecation policy (no symbol removed without a `DeprecationWarning` for at least one minor release), and ships a Keep a Changelog history. The contract is enforced by building the wheel and importing it from a *clean*, network-free install into a fresh interpreter that cannot see the source checkout—then verifying a real shape bug end-to-end.

**Continuous integration that pays only for the diff.** The GitHub Action annotates the exact offending lines of a pull-request diff and emits SARIF for code scanning. A *changed-only* mode restricts verification to the `*.py` models a PR actually touches (`git diff --diff-filter=ACMRT base...head`), ignoring deletions and non-Python files and degrading gracefully to a full scan when the base ref is unobtainable (shallow clone, fork, non-`main` default branch). A baseline file plus inline `# tensorguard: ignore` suppressions enable incremental adoption on legacy repositories, so only *new* findings can fail the gate.

**The test suite as a verification gate.** The `pytest` plugin (`pytest --tensorguard`) discovers the project modules the suite *actually imports*—snapshotting `sys.modules` at session start and collecting the modules first imported afterwards whose source lives under the project root—and verifies that exercised code, not a blind directory walk. An imported buggy model fails the session while an unimported "orphan" file in the same tree is left alone; an explicit path or a whole-tree mode remain available.

**Inline feedback in the editor.** A real Language Server (`python -m src.lsp_server`, JSON-RPC over stdio with byte-accurate framing) analyses the *unsaved editor buffer* on every change, publishing shape/device/dtype/phase/gradient squiggles, hover shapes drawn from the inference chain, and quick-fixes whose workspace edit reproduces the mechanical autofix—and clearing the diagnostics the instant the code becomes correct. A VS Code extension is a thin language client over it, so the editor is a transport rather than a re-implementation.

**Verification inside the compile pipeline.** For users who reach for `torch.compile`, `make_tensorguard_backend` is a Dynamo backend that runs verification as a guard: Dynamo hands it the captured `GraphModule` and example inputs, the backend verifies the module's real source, then delegates a clean model to an inner compiler whose output reproduces eager—or raises `TensorGuardViolation` (carrying the structured findings) on a buggy one, surfacing through the compile pipeline's exception chain. The complementary `guarded_compile` pre-pass blocks a bug *before* compilation even on interpreters where Dynamo is unavailable, so the same shape error that would otherwise appear as an opaque guard failure or a deep inductor traceback is reported at the model's own source line.

**Gating the training and export lifecycle.** The same pre-pass guards the two other points where a model leaves the user's hands. `prepare_verified` wraps `accelerate.Accelerator.prepare`—the choke point every model passes through before distributed/mixed-precision training—running verification as the *first* side effect on the user's own `nn.Module`, so a real bug raises before the model is wrapped or moved (otherwise it surfaces mid-epoch inside an AMP/DDP shim whose traceback no longer points at `forward`); the wrapper preserves `prepare`'s exact bare-object-vs-tuple return contract. `guarded_onnx_export` gates `torch.onnx.export`, verifying *before* anything is written to the export sink—inferring the verified input shapes from the example arguments so the shape that is checked is the shape that is exported—turning a malformed graph or an opaque tracer error into one actionable `TensorGuardViolation`. Lightning and Hugging Face `Trainer` callbacks verify the model at `on_fit_start` / `on_train_begin`. Each gate is closed-loop tested against the real framework: a live CPU `Accelerator` (clean prepares and runs; a spy confirms a buggy model never reaches `prepare`), the real `torch.onnx` exporter (a clean model yields a proto that passes `onnx.checker`; a buggy one leaves the sink byte-for-byte empty), and a real `pytorch_lightning.Trainer.fit`.

**Whole-lifecycle parity and a community scoreboard.** The gate set now spans every point a model leaves the user's hands. `guarded_from_pretrained` verifies a returned `PreTrainedModel` before the Hugging Face loader hands it back—a gated clean checkpoint then trains for real under a genuine `transformers.Trainer`, while a misbuilt one raises before it can reach a pipeline. `verify_exported_program` and `guarded_aot_package` bring the `torch.export` and AOTInductor packaging paths to parity with the ONNX gate (verification is the first side effect; shapes are inferred from the example arguments; no `.pt2` is written on a bug), and `guarded_onnx_export` adds an `onnx.checker.check_model` post-export assertion so a structurally invalid graph fails at export rather than downstream load. To keep the empirical claims honest *after* publication, an open leaderboard re-scores external tool submissions in CI: a contributor commits only raw per-case verdicts on the frozen, hash-pinned corpus, a validator rejects unknown ids, malformed verdicts, and any self-reported metric, and the harness recomputes recall/precision/F1 itself—so a tool that claims perfect recall but abstains everywhere is scored at its true recall of zero. Finally, `upstream_hook.install()` grafts the proposed `torch.nn.utils.verify_module` / `attach_verifier` / `torch.nn.verifiable` surface onto the live namespace with no core changes—a reversible, idempotent preview of the RFC's Phase-1 API, exercised directly against real `torch`.

**One engine, many doors.** Table 15 summarizes the developer-facing entry points; crucially, all of them route through the same verified core, so no surface can drift from the soundness guarantees proved in Section 7. Every integration above is regression-tested against a real wheel, a real git repository, a real subprocess `pytest` run, a real LSP handshake, and a real FX graph, respectively.

# 11 Threats to Validity

**Internal validity.** The headline recall/precision could in principle reflect tuning to the development corpus. We mitigate this with the pre-registered blind split of Section 6.3 (manifest hash fixed before evaluation; H1–H3 confirmed on held-out data) and with multiple-comparison correction on every comparative claim. The abstention accounting is explicit: `sound` mode converts every out-of-fragment construct into `UNKNOWN` rather than a silent verdict, so a missed bug surfaces as an abstention rather than an unsound `SAFE`.

**Construct validity.** The strongest threat is the *conformance gap*: a transfer function could mis-model PyTorch and the proofs would faithfully certify the wrong semantics. This is precisely why the Lean soundness modules are each paired with a test that runs the *live* verifier, real Z3, and the real torch dispatcher (Section 7); the remaining gap is the per-operator bit-level faithfulness of the shape transfer functions, bounded by property-based differential testing against the live dispatcher.

**External validity.** Results may not transfer to model families outside the corpus. The corpus is stratified by bug class and drawn from real public repositories, the false-positive stress corpus uses clean real models, and the cross-version/cross-interpreter matrices (Section 6.4) show the verdict is invariant to the torch and Python versions a practitioner runs. The phase domain is reported as diagnostic-only to avoid overclaiming.

**Reproducibility.** Wall-clock numbers are machine-dependent and are regenerated but not byte-compared; all categorical verdicts are content-addressed and reproduce bit-for-bit under the pinned capsule (Section 6.6).

# 12 Conclusion

We have presented TensorGuard, a static multi-property verifier for PyTorch that combines five domain-specific SMT theories—shape, device, phase, stride, and permutation—to catch cross-cutting bugs that no existing tool detects. On a 52-model benchmark of multi-theory bugs sourced from real-world issues (HuggingFace #13666, torchvision ResNet, detectron2, YOLOv5, mmdetection), TensorGuard achieves F1 = 0.625 with perfect precision. The baseline rows in this comparison should be read as capability lower bounds, not full per-case reruns: TorchScript, mypy, jaxtyping, and PyTEA lack the device/phase reasoning that defines these cross-cutting bug classes. The tool supports 142 operator transfer functions, covers real-world architectures (ResNet, VGG, MobileNetV2, EfficientNet, DenseNet), and produces parametric safety certificates via IC3/PDR over symbolic dimensions. The repository also includes a compiled, `sorry`-free Lean 4 formalization (∼3,800 lines, 210 theorems/lemmas, *all* machine-audited to trusted kernel axioms) covering the operator transfer functions, the reduced-product abstraction, CEGAR termination, the two known-unsoundness gaps, the device/dtype/phase/gradient transfer functions, and the faithfulness of the Z3 encoding the solver runs.

The primary limitation is the conformance gap between the mechanized specification and the Python implementation. The remaining false negatives on cross-cutting benchmarks (Recall = 0.455) stem from device-only bugs where shapes match perfectly; extending the device theory to track cross-device operations through `@` and `expand` is future work. Additional directions include closing the conformance gap via verified extraction, extending the operator registry toward full PyTorch coverage, and integrating TensorGuard into CI/CD pipelines as a pre-merge gate for multi-property safety.

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# A Complete Typing Rules

This appendix presents the complete set of typing rules for all 142 operator transfer functions. We group them by category.

## A.1 Convolution Operators

**T-Conv1d.**

$$\frac{\Gamma \vdash x : \{t \mid ndim(t) = 3 \land dim_1(t) = c_{\text{in}}\}}{\Gamma \vdash \texttt{Conv1d}(c_{\text{in}}, c_{\text{out}}, k)(x) : \{t' \mid dim_0(t') = dim_0(x) \land dim_1(t') = c_{\text{out}} \land dim_2(t') = \lfloor \frac{dim_2(x) + 2p - d(k-1) - 1}{st} \rfloor + 1\}} \text{ T-C}$$

**T-Conv3d.**

$$\frac{\Gamma \vdash x : \{t \mid ndim(t) = 5 \land dim_1(t) = c_{\text{in}}\}}{\Gamma \vdash \texttt{Conv3d}(c_{\text{in}}, c_{\text{out}}, k)(x) : \{t' \mid dim_0(t') = dim_0(x) \land dim_1(t') = c_{\text{out}} \land \forall i \in \{2,3,4\}.\ dim_i(t') = \lfloor \frac{dim_i(x) + 2p_i - d_i(k_i - 1)}{st_i}$$

**T-ConvTranspose2d.**

$$\frac{\Gamma \vdash x : \{t \mid ndim(t) = 4 \land dim_1(t) = c_{\text{in}}\}}{\Gamma \vdash \texttt{ConvTranspose2d}(c_{\text{in}}, c_{\text{out}}, k)(x) : \{t' \mid dim_0(t') = dim_0(x) \land dim_1(t') = c_{\text{out}} \land dim_i(t') = (dim_i(x) - 1) \cdot st_i - 2p_i + d_i(k$$

## A.2 Normalization Operators

**T-BatchNorm.**

$$\frac{\Gamma \vdash x : \{t \mid dim_1(t) = n\}}{\Gamma \vdash \texttt{BatchNorm}(n)(x) : \{t' \mid shape(t') = shape(x)\}}\ \text{T-BatchNorm}$$

**T-GroupNorm.**

$$\frac{\Gamma \vdash x : \{t \mid dim_1(t) = c \land c \bmod g = 0\} \quad g \in \mathbb{Z}_{>0}}{\Gamma \vdash \texttt{GroupNorm}(g, c)(x) : \{t' \mid shape(t') = shape(x)\}}\ \text{T-GroupNorm}$$

## A.3 Structural Operators

**T-Flatten.**

$$\frac{\Gamma \vdash x : \{t \mid shape(t) = s\} \qquad 0 \le d_s \le d_e < ndim(x)}{\Gamma \vdash \texttt{flatten}(x, d_s, d_e) : \{t' \mid shape(t') = s_{[0:d_s]} \parallel [\prod_{i=d_s}^{d_e} s_i] \parallel s_{[d_e+1:]}\}}\ \text{T-Flatten}$$

**T-Transpose.**

$$\frac{\Gamma \vdash x : \{t \mid shape(t) = s\} \qquad 0 \le d_0, d_1 < ndim(x)}{\Gamma \vdash \texttt{transpose}(x, d_0, d_1) : \{t' \mid dim_{d_0}(t') = s_{d_1} \land dim_{d_1}(t') = s_{d_0} \land \forall k \notin \{d_0, d_1\}.\ dim_k(t') = s_k\}}\ \text{T-Transpose}$$

**T-Split.**

$$\frac{\Gamma \vdash x : \{t \mid shape(t) = s\} \qquad s_d \bmod n = 0}{\Gamma \vdash \texttt{split}(x, s_d/n, d) : [t'_1, \ldots, t'_n] \text{ where } \forall i.\ dim_d(t'_i) = s_d/n}\ \text{T-Split}$$

## A.4 Pooling Operators

**T-MaxPool2d.**

$$\frac{\Gamma \vdash x : \{t \mid ndim(t) = 4\}}{\Gamma \vdash \texttt{MaxPool2d}(k, st, p)(x) : \{t' \mid dim_{0,1}(t') = dim_{0,1}(x) \land \forall i \in \{2,3\}.\ dim_i(t') = \lfloor \frac{dim_i(x) + 2p_i - k_i}{st_i} \rfloor + 1\}}\ \text{T-MaxPool2d}$$

**T-AdaptiveAvgPool2d.**

$$\frac{\Gamma \vdash x : \{t \mid ndim(t) = 4\}}{\Gamma \vdash \texttt{AdaptiveAvgPool2d}(h', w')(x) : \{t' \mid dim_{0,1}(t') = dim_{0,1}(x) \land dim_2(t') = h' \land dim_3(t') = w'\}}\ \text{T-AdaptiveAvgPool2d}$$

## A.5 Attention Operators

**T-MultiheadAttention.**

$$\frac{\Gamma \vdash q : \{t \mid shape(t) = (L, N, E)\} \Gamma \vdash k : \{t \mid shape(t) = (S, N, E_k)\} \Gamma \vdash v : \{t \mid shape(t) = (S, N, E_v)\} E \bmod h = 0}{\Gamma \vdash \texttt{MHA}(E, h)(q, k, v) : \{t' \mid shape(t') = (L, N, E)\}}\ \text{T-MHA}$$

### A.6 Recurrence Operators

**T-LSTM.**

$$\frac{\Gamma \vdash x : \{t \mid ndim(t) = 3 \land dim_2(t) = f_{\text{in}}\}}{\Gamma \vdash \texttt{LSTM}(f_{\text{in}}, h, L, bi)(x) : \{(o, (h_n, c_n)) \mid dim_2(o) = h \cdot (1 + bi)\}} \text{ T-LSTM}$$

## B Full Operator List

The 142 operators are organized into the following categories. Each entry shows the operator name and the SMT fragment of its generated constraints.

**Convolution:** Conv1d, Conv2d, Conv3d, ConvTranspose1d, ConvTranspose2d, ConvTranspose3d (all QF_LIA)

**Normalization:** BatchNorm1d, BatchNorm2d, BatchNorm3d, LayerNorm, GroupNorm, InstanceNorm1d, InstanceNorm2d, InstanceNorm3d (all QF_LIA)

**Attention:** MultiheadAttention, ScaledDotProductAttention, RoPE (QF_LIA)

**Recurrence:** LSTM, GRU, RNN, LSTMCell, GRUCell, RNNCell (QF_LIA)

**Pooling:** MaxPool1d, MaxPool2d, MaxPool3d, AvgPool1d, AvgPool2d, AvgPool3d, AdaptiveMaxPool1d, AdaptiveMaxPool2d, AdaptiveMaxPool3d, AdaptiveAvgPool1d, AdaptiveAvgPool2d, AdaptiveAvgPool3d (QF_LIA)

**Activation:** ReLU, LeakyReLU, PReLU, ELU, SELU, GELU, Sigmoid, Tanh, Softmax, LogSoftmax, Hardswish, Mish, SiLU (shape-preserving)

**Padding:** ZeroPad1d, ZeroPad2d, ConstantPad1d, ConstantPad2d, ConstantPad3d, ReflectionPad1d, ReflectionPad2d, ReplicationPad1d, ReplicationPad2d, ReplicationPad3d (QF_LIA)

**Loss:** CrossEntropyLoss, MSELoss, NLLLoss, BCELoss, BCEWithLogitsLoss, L1Loss, SmoothL1Loss, KLDivLoss, HuberLoss (QF_LIA)

**Embedding:** Embedding, EmbeddingBag (QF_LIA)

**Structural:** reshape, view, flatten, unflatten, cat, stack, split, chunk, squeeze, unsqueeze, transpose, permute, contiguous, expand, repeat, narrow, select, index_select, gather, scatter, einops rearrange/repeat/reduce, PixelShuffle (reshape/flatten/PixelShuffle/repeat: QF_NIA; others: QF_LIA)

## C Proof Sketches for Soundness Conjectures

### C.1 Type Soundness (Conjecture 1)

*Extended proof sketch.* We proceed by induction on the typing derivation $\Gamma \vdash e : \tau$.

**Base case (T-Var):** If $e = x$ and $\Gamma(x) = \tau$, then $\sigma \models \Gamma$ implies $\sigma(x) \models \tau$ by definition.

**Inductive case (T-Linear):** Assume $\Gamma \vdash x : \{t \mid dim_{-1}(t) = f_{\text{in}}\}$. By the induction hypothesis, $\sigma(x)$ has last dimension $f_{\text{in}}$. The PyTorch `nn.Linear` documentation specifies that the output shape is $shape(x)_{[0:-1]} \parallel [f_{\text{out}}]$ when the input's last dimension equals $f_{\text{in}}$. Since the precondition holds, no `RuntimeError` is raised, and the output satisfies the postcondition. The argument is analogous for all other rules.

**Theory combination:** Cross-theory constraints (e.g., device compatibility implying shape compatibility) are sound by Theorem 1 (Lean-proved).

**Gap:** The proof sketch assumes that each transfer function faithfully models the corresponding PyTorch operator. This assumption is not mechanized; it is validated by property-based testing against the PyTorch runtime. □

### C.2 Stride Theory Convexity

*Proof sketch.* $\mathcal{T}_{stride}$ generates constraints in QF_LIA over $\mathbb{Z}_{\geq 1}$. QF_LIA is convex: if $\varphi \models e_1 = e_2 \vee e_3 = e_4$, then $\varphi \models e_1 = e_2$ or $\varphi \models e_3 = e_4$. This follows from the fact that the solution set of a conjunction of linear inequalities over the integers is a finite union of lattice points in a convex polytope, and equality disjunctions over such sets reduce to individual equalities [7].

The NIA subfragment arising from reshape constraints uses at most one symbolic factor per product term, restricting to a semi-linear fragment where convexity is preserved. Full nonlinear arithmetic would break convexity; our restriction to semi-linear terms avoids this. □

## D Machine-Checked Faithfulness Theorems

For reference, we state the central faithfulness and transfer results added in this work (all `sorry`-free, depending only on `propext`). Let $\mathrm{Sat}(c_a, c_b)$ denote satisfiability of the pinned same-value constraint the verifier emits, and let devBug, etc., be the abstract algebra predicates of `DeviceDtype.lean`.

- **Enum-encoding faithfulness** (`SmtEncoding.lean`): $\mathrm{Sat}(c_a, c_b) \iff c_a = c_b$, hence $\neg\mathrm{Sat}(c_a, c_b) \iff c_a \neq c_b$; and for known devices $\neg\mathrm{Sat}(c_a, c_b) \iff \mathrm{devBug}(c_a, c_b)$. The same UNSAT characterization holds for the phase sort and the gradient Boolean.
- **Cross-domain faithfulness** (`CrossDomain.lean`): a transfer op's constraint set is satisfiable $\iff$ `pre`=`post` (shape preserved, device free); a non-transfer op's $\iff$ $d_{\mathrm{pre}} = d_{\mathrm{post}}$ (device preserved, shape free).
- **Dtype promotion semilattice** (`DeviceDtype.lean`): dtPromote is commutative, idempotent, associative, and $\top$-absorbing; hence the elementwise dtype transfer never flags ($\mathrm{dtElementwiseBug} \equiv$ `false`).
- **Gradient-flow chain transfer** (`GradFlow.lean`): gradRun is compositional ($\mathrm{gradRun}\, b\, (xs +\!+ ys) = \mathrm{gradRun}\, (\mathrm{gradRun}\, b\, xs)\, ys$); `detach`/`noGrad` are absorbing; and on the reattach-free fragment the outgoing bit is `true` iff the input required grad and no reset intervened.
- **Device-placement chain** (`DevicePlacement.lean`): the device tag is last-move-wins ($\mathrm{devRun}\, d_0\, (xs +\!+ [op])$ is *op*'s target whenever *op* is a move, independent of $d_0$), preserved on the move-free fragment, and a binary op is valid iff its operands share a tag ($\mathrm{binValid}(a, b) \iff a = b$) — a cross-device pair is always flagged, a same-device pair never falsely flagged.
- **Train/eval phase chain** (`PhaseFlow.lean`): the `training` bit is last-setter-wins and preserved on the setter-free fragment; the run is compositional, matching real `nn.Module` mode propagation (parent and children).
- **Reduction rank chain** (`RankTransfer.lean`): the rank transfer is compositional and monotone non-increasing ($\mathrm{rankRun}\, r_0\, xs \leq r_0$), with closed form $\mathrm{rankRun}\, r_0\, xs = r_0 - \#\{$`keepdim=False`$\}$ (truncated), so `keepdim=True` preserves rank and the verifier never invents a dimension.
- **Reduced-product lifting** (`DeviceDtype.lean`): the product over the four scalar algebras flags iff some component flags, and inherits both no-false-positive and refutation soundness.

Each item is gated by a Python test that reproduces the predicted SAT/UNSAT/verdict on the *real* solver or the *real* torch dispatcher; the whole library's 371 theorems are audited `sorry`-free by `test_lean_whole_pipeline_audit.py`.

## E Trusted Computing Base Summary

The trusted computing base (TCB) of TensorGuard comprises the following components, listed in decreasing order of trust required:

1. **Z3 SMT solver.** We trust Z3's decision procedures for QF_LIA, QF_NIA, and QF_UF. Z3 is widely used and extensively tested, but bugs are possible.

2. **Lean 4 proofs.** The Lean 4 formalization is compiled (`lake build`) and `sorry`-free, and its headline soundness results are machine-audited to rest only on the trusted Lean kernel axioms. The Lean checker is not invoked at TensorGuard runtime, so it constrains the *specification*; the conformance gap to the Python implementation is closed test-by-test by the live cross-checks (Section 7).

3. **Python implementation.** The TensorGuard Python code (graph extraction, transfer functions, constraint generation) is *not* mechanically verified. The conformance gap between the Lean specification and the Python implementation is mitigated by:
   - Live cross-checks that assert the verifier reproduces each Lean decision function's predictions on real `nn.Module`s.
   - Property-based testing (Hypothesis) exercising mechanized theorem statements on the implementation.
   - Extensive unit and integration tests (3,709 tests total).
   - Evaluation on real-world models providing empirical confidence.

4. **Transfer function faithfulness.** Each of the 142 transfer functions is assumed to faithfully model the corresponding PyTorch operator. This is validated by property-based testing against the PyTorch runtime but not mechanically proved.

5. **Graph extraction correctness.** The AST-based and FX-based graph extractors are assumed to correctly reconstruct the computation graph from source code. Dynamic dispatch, monkey-patching, and metaclass magic can cause extraction failures (reported as warnings, not silent misses).

# F Reproducibility Footprint

TensorGuard ships not a single benchmark but a layered evaluation and reproducibility suite, all in-tree and regenerable. Table 16 summarizes the deployed assets; each row is materialized from committed code and artifacts (no network or GPU required for the headline rows), and every headline number cited in this paper is covered by the single-harness numeric-claims audit (`reproducibility/audit_numeric_claims.py`), which classifies each as VERIFIED or flags a regime/environment qualifier.

The design goal is that a reviewer can move from any claim in the text to the exact line of code or artifact that produces it, and rerun it—the "credibility" half of the contribution, as load-bearing as the verification engine itself.

Table 12: Machine-checked soundness modules added in this work (all `sorry`-free, `propext`-only, each gated by a Python conformance test against real code).

| Lean module | Cross-check | Guarantee |
|---|---|---|
| `DeviceDtype.lean` | live verifier + torch | 4 algebras: no-FP + refutation-sound; reduced product |
| `SmtEncoding.lean` | real Z3 | device/phase/grad enum encoding UNSAT ⇔ endpoints differ |
| `CrossDomain.lean` | real Z3 | transfer preserves shape / non-transfer preserves device, exactly |
| `GradFlow.lean` | real autograd | `requires_grad` chain transfer; compositional, absorbing resets |
| `DevicePlacement.lean` | real torch (cpu/mps) | device-tag chain; last-move-wins; binary op valid ⇔ same device |
| `PhaseFlow.lean` | real `nn.Module` | train/eval bit chain; last-setter-wins, child propagation |
| `RankTransfer.lean` | real torch `sum` | reduction rank chain; monotone, closed-form rank drop |
| `DtypePromoteChain.lean` | `torch.promote_types` | promotion fold; upper bound of every operand, order-independent |
| `BroadcastChain.lean` | `torch.broadcast_shapes` | per-dim broadcast fold; `1` identity, compat = max, flags iff incompatible |
| `ContigFlow.lean` | real torch strides | contiguity bit chain; `.t()` involution, `.contiguous()` history-erasing |
| `SmtEncoding.lean` (dtype) | real Z3 + torch matmul | matmul dtype encoding UNSAT ⇔ dtypes differ ⇔ `dtMatmulBug` |
| `Flatten.lean` | torch + `_propagate_flatten` | numel-preserving span collapse; rank law; full flatten = [numel] |
| `CatRule.lean` | real `torch.cat` | admissible ⇔ non-axis dims match; axis/numel additive |
| `EmbeddingRule.lean` | real `nn.Embedding` | rank +1, trailing = `embedding_dim`; index guard iff in range |
| `ReshapeInfer.lean` | torch + `compute_reshape_shape` | `-1`-inference = numel/$\prod known$; valid iff it divides |
| `LinearRule.lean` | real `nn.Linear` | rank preserved, last dim = `out_features`; `in_features` guard |
| `Conv2d/Conv1d.lean` | real `nn.Conv`$\{1,2\}$`d` | spatial formula: identity, monotone, padded-input bound, positive-output guard |
| `Pool2d.lean` | real `nn.MaxPool2d/AvgPool2d` | pooling spatial formula; channel-preserving |
| `ConvTranspose.lean` | real `nn.ConvTranspose1d` | length formula; upsampling lower bound (never shrinks) |
| `LayerNorm/BatchNormRule` | real `nn.{Layer,Batch}Norm` | shape-preserving; suffix / feature-count guard |
| `PixelShuffle.lean` | real `nn.PixelShuffle` | numel-preserving; admissible ⇔ $r^2 \mid C_{\text{in}}$, $C_{\text{out}} = C_{\text{in}}/r^2$ |
| `AdaptivePool.lean` | real `nn.AdaptiveAvgPool2d` | spatial = target for any input; idempotent |
| `Unflatten.lean` | real `nn.Unflatten` | numel-preserving inverse of flatten; size guard |

Table 13: Chain-transfer conformance: each proved per-`forward` transfer is replayed exhaustively (all chains of length $\leq 3$) against the live framework and the verdict checked against the Lean prediction.

| Proved transfer | Replayed against | Live invariant checked |
|---|---|---|
| device placement | real `cpu`/`mps` tensors | `.device.type`; cross-device `+` raises iff ¬binValid |
| train/eval phase | real `nn.Sequential` | `.training` on parent *and* every child |
| reduction rank | real `torch.sum` | `out.dim()` = rankRun |
| gradient flow | real autograd tensor | `out.requires_grad` = gradRun |

| **Confidence** | **Count** | **Representative operators** |
|---|---|---|
| Complete | 75 | pointwise activations, `reshape`, `permute`, `cat`, broadcasting binops |
| Sound | 36 | `matmul`, `bmm`, `einsum`-free reductions, `fft.*`, `gather` |
| Heuristic | 6 | `einsum`, `unique`, `multinomial`, `linalg.{qr,svd,solve}` |
| **Total** | **117** | |

Table 14: Per-operator soundness coverage, exported by `tensorguard operator-confidence`. The complete/sound split is honored at run time: `sound` mode abstains on heuristic operators rather than emitting a verdict.

| **Entry point** | **Purpose** |
|---|---|
| `analyze` / `analyze-package` | verify a file/module or a whole installed package (parallel workers) |
| `verify` | shape/device/phase/grad checks with explicit input shape and CEGAR depth |
| `ci-check` | baseline diff with `--max-new-bugs` budget; SARIF output for code scanning |
| `watch` | incremental re-verification on save (vim/emacs/vscode diagnostics) |
| `server` | long-lived analysis server over stdio or TCP |
| `report` / `export` | render HTML/SARIF/JSON; emit `.pyi/` `.d.ts` shape stubs |
| `diff` | semantic diff between two analysis runs |
| `operator-confidence` | query the soundness tag of any operator |
| `pre-commit` hook | block commits that introduce new bugs |
| `pytest` plugin | assert shape safety from within a test suite |
| `guarded_compile` / Dynamo backend | verify inside the `torch.compile` pipeline |
| `guarded_onnx_export` | gate `torch.onnx.export` on a clean verdict |
| `prepare_verified` | gate `accelerate.Accelerator.prepare` |
| Lightning / HF `Trainer` callbacks | verify the model before the first training step |

Table 15: The integration surface: one engine, many developer-facing entry points, all sharing the verified core.

Table 16: Deployed verification and evaluation assets (selected).

| Asset | What it establishes |
|---|---|
| Cross-cutting suite (52 models) | multi-theory bugs no shape-only tool catches; precision 1.0, 0 FP |
| PyTEA fragment-fair set ($N$=34) | shape-bug parity vs. a published checker; McNemar exact $p$=0.0156 |
| Frozen ground-truth corpus | 16 SHA-pinned real `nn.Module`s (8 clean / 8 buggy), verdicts replayed in eager torch |
| Per-domain ablation corpus | device and gradient domains each refute bugs the shape view misses |
| Operator confidence table | all 117 transfer functions tagged complete/-sound/heuristic, code↔table synced |
| Lean 4 library | 371 `sorry`-free theorems, kernel-axiom audited, drift-proof |
| Conformance + chain-transfer tests | Lean predictions replayed on real Z3, autograd, `cpu`/`mps`, `nn.Module`s |
| Test suite | 3,709 unit/integration/property tests gating the above |